SPACE APPS 2026

PROJECT BUILD GUIDE

Interplanetary Survival Map

Quanta Buddies

A practical engineering, scientific-validation, and competition-readiness guide for building a reliable Mars surface-exploration planning tool.

CORE PRINCIPLE
Build a scientifically transparent, offline-capable decision-support tool. Use NASA data as evidence, deterministic code for calculations, and AI only where it adds explainable value.

1. Mission and Product Goal

Build an interactive planning environment that helps users explore Mars surface regions and assess candidate traverse routes using available planetary data. The application should connect terrain, science objectives, hazards, route alternatives, and mission-planning assumptions in one coherent workflow.

- Primary user: a future surface-exploration planner or astronaut-support team.
- Primary outcome: a transparent Marswalk planning aid—not a certified mission navigation or life-support system.
- Coverage: all 10 regions already supported by the project. Jezero may be the main demonstration region, but region-aware logic must work everywhere.
- Competition goal: demonstrate a clear connection to the challenge, meaningful NASA data use, scientific care, usability, teamwork, and a reproducible four-minute story.

2. Architecture Principles

2.1 Recommended system structure

- Frontend: retain the existing React + TypeScript + Vite interface, CesiumJS/Leaflet views, and current visual design.
- Application/API layer: expose stable interfaces for terrain, hazards, science targets, route planning, and mission summaries.
- Data layer: store prepared NASA-derived datasets locally in documented formats such as GeoTIFF/COG, Zarr, Parquet, JSON fixtures, or PMTiles where appropriate.
- Compute layer: keep geometry, interpolation, slope, distance, route cost, and energy/time estimates in deterministic, testable functions.
- AI/agent layer: use AI only to explain evidence, summarize context, or help compare options. AI must not replace scientific calculations.
- Acquisition layer: collect and prepare data before the demo or through a controlled background workflow; the demo should not depend on live NASA downloads.

2.2 Separation rules

- The frontend should not call NASA services directly for every map interaction.
- The demo API should read from prepared local data/cache rather than depending on live external services.
- Data acquisition, scientific processing, route computation, and AI explanation should remain separate responsibilities.
- Do not add FastAPI, MCP servers, agent orchestration, or other infrastructure unless a real requirement justifies the added complexity.

3. NASA Data Acquisition and Provenance

Every scientific layer must have a traceable origin. A dataset name alone is not sufficient evidence.

3.1 Candidate Mars data sources

Dataset family	Appropriate use and required caution
MOLA / MGS MEGDR	Global elevation context and regional terrain baseline; document product ID, grid spacing, bounds, and vertical reference.
HiRISE / CTX DTMs	Higher-resolution local terrain where coverage exists; never imply uniform coverage across all 10 regions.
THEMIS	Thermal-infrared context; identify product type, resolution, acquisition time, and processing.
CRISM	Spectral/mineralogical context; explain whether a layer is a browse product, calibrated product, or derived interpretation.
SHARAD / SWIM	Subsurface or ice-related context; clearly distinguish vertical from horizontal resolution and modelled interpretations.
MEDA / InSight archives	In-situ environmental or seismic observations at their actual landing-site locations; do not treat them as live, global Mars weather.
Mars Trek / PDS	Discovery, visualization, and source-product access; cite the exact

underlying product where possible.

3.2 Required provenance record

- Dataset/product title and unique product ID.
- Original source URL and archive/provider.
- Spatial coverage and coordinate reference system.
- Native resolution and the resolution used by the application.
- Acquisition date/time or mission/sol where relevant.
- Processing steps: reprojection, cropping, resampling, interpolation, filtering, and downsampling.
- Known gaps, uncertainty, quality flags, and licensing/usage conditions.
- A clear label: VERIFIED, PARTIALLY VERIFIED, RASTER-DERIVED, SIMULATED, or UNAVAILABLE.

3.3 Ten-region coverage rule

Maintain a region-by-region provenance table. For each of the 10 regions, record which source products actually cover the region. If a region uses a derived, interpolated, synthetic, or placeholder grid, label it honestly. Do not describe all regional matrices as authentic HiRISE/CTX data unless the source coverage and processing chain are documented.

4. Offline-First and Reliable Demo Operation

- Pre-fetch all datasets, tiles, map styles, and demo assets required for the chosen demonstration.
- Provide a deliberate fallback order: live source when available → local cache → clearly labelled fixture/demo data.
- Support an OFFLINE=1 or equivalent mode that disables network-dependent calls and uses cached content.
- Do not let a quota limit, API outage, or internet interruption break the core map and route workflow.
- Show a visible data-status label so users can distinguish live, cached, archived, simulated, and unavailable information.
- Rehearse the complete demonstration with Wi-Fi disabled.

5. Deterministic Scientific and Geospatial Computation

5.1 Terrain and map calculations

- Use the selected region consistently for elevation lookup, map rendering, hazards, science targets, routing, and emergency-return calculations.
- Verify coordinate order, coordinate reference system, bounds, no-data handling, and unit conversions.
- Document how raster grids are sampled and how bilinear interpolation or other resampling affects results.
- Calculate slope from a documented elevation surface and state the grid resolution used.
- Do not imply that a coarse global DEM can resolve small rocks, cracks, or other sub-grid obstacles.

5.2 Route planning

- Support user-selected start and destination points in every supported region.
- Sample terrain along the complete route and produce an elevation/slope profile.
- Calculate distance and route cost with documented units and assumptions.
- Represent hazards and science targets using region-specific data.
- Compare route alternatives using transparent criteria such as terrain cost, hazard exposure, distance, and science value.
- Provide an emergency/return alternative where the available data support it.
- Test missing data, blocked routes, invalid coordinates, region switching, and disconnected map areas.

5.3 Human astronaut versus rover assumptions

The challenge concerns human surface exploration. Rover slope thresholds, mass, power, traction, and tipping limits must not be presented as astronaut EVA limits. If a rover-support mode exists, label it separately. Human route time, energy, suit constraints,

and safe operating limits must be explicitly described as assumptions or estimates unless supported by an appropriate validated model.

6. AI and Agentic Features

- AI may summarize verified datasets, explain route trade-offs, describe uncertainty, and produce contextual suggestions.
- AI must not invent NASA citations, product IDs, measurements, or unavailable observations.
- The AI response should identify the evidence used and distinguish fact, inference, and simulation.
- Keep route geometry, slope, distance, energy/time calculations, and scoring in deterministic code.
- If an AI service is unavailable or its quota is exhausted, core functionality must continue with a clearly labelled fallback.
- Use MCP tools or multi-agent orchestration only when they provide a demonstrable capability; do not add them for appearance.

7. Safety, Risk, and Scientific Transparency

- Describe the application as a research prototype or decision-support demonstrator, not a flight-qualified or mission-certified system.
- Show assumptions and uncertainty beside estimates—not hidden in documentation alone.
- Do not label a site a verified habitation site based on a heuristic score.
- Avoid unsupported claims about local radiation, pressure, boulder abundance, ice depth, or slope.
- Distinguish measured observations, archived observations, model outputs, derived layers, analytical estimates, and simulations.
- Use conservative wording such as 'candidate region for further assessment' when evidence is regional or incomplete.

8. User Experience and Core Workflow

A judge or first-time user should be able to complete the main workflow without a long explanation:

1. Select one of the 10 Mars regions.
2. Inspect the available terrain and NASA-derived context layers.
3. Choose a science objective and set start/destination waypoints.
4. Generate and compare route alternatives.
5. Inspect distance, elevation, slope, hazards, and estimated mission costs.
6. Review the evidence and uncertainty behind the AI explanation.
7. Open an emergency/contingency route and understand its assumptions.

9. Testing and Quality Gates

Quality gate	Acceptance check
Region integration	All 10 regions load the correct terrain, hazards, science targets, and map bounds.
Routing	Start/destination, route sampling, alternatives, hazard avoidance, and emergency return work per region.
Data integrity	No silent fallback to Jezero; no-data and missing-source cases are handled visibly.
Scientific calculations	Units, coordinate transforms, interpolation, slope, and distance are tested against known fixtures.
AI behaviour	No fabricated sources; unavailable data and uncertainty are communicated.
Offline behaviour	The full core workflow works without network access.
Build quality	TypeScript, lint, automated tests, and production build pass.

A passing test suite confirms tested software behaviour; it does not independently prove that a NASA dataset is authentic or that a scientific assumption is correct. Scientific validation requires source evidence and appropriate comparison.

10. Documentation, Reproducibility, and Responsible AI

- Maintain a README with project purpose, challenge mapping, setup instructions, architecture, data sources, and limitations.
- Document data preparation commands and the steps needed to reproduce derived layers.
- Include small test fixtures so core calculations can be reproduced without downloading large datasets.
- Use a clear open-source licence, such as Apache-2.0, only after confirming the project is eligible to use it and all included assets are compatible.
- Add an AI-use disclosure explaining which features use AI, what the AI receives, and which calculations remain deterministic.
- Record known limitations and do not hide incomplete features behind polished interface labels.

11. Competition Presentation: 240-Second Demo

Time	What to demonstrate
0:00–0:25 | The problem	Explain why combining terrain, science targets, and hazards matters for planning a Mars surface excursion.
0:25–0:55 | NASA data	Show the selected region and identify the real source layers, their resolution, and any coverage limitations.
0:55–1:35 | Plan a traverse	Select a start, destination, and science objective; generate route alternatives.
1:35–2:15 | Explain the trade-off	Compare terrain, slope, hazards, distance, and science value. Show assumptions and uncertainty.
2:15–2:45 | Contingency	Demonstrate an emergency/return option and explain what data and model assumptions it uses.
2:45–3:20 | AI role	Show how AI explains evidence and route trade-offs without performing or inventing scientific calculations.
3:20–4:00 | Impact and honesty	State what works, what is simulated or incomplete, and how the project can be extended.

12. Final Implementation Checklist

☐ The current React application and its working interface are preserved.

☐ All 10 regions are genuinely region-aware throughout the application.

☐ Every dataset has a source/provenance record and honest status label.

☐ Regional DEM coverage and resolution are documented; derived grids are not misrepresented.

☐ Terrain, routing, and mission estimates are deterministic and testable.

☐ Astronaut assumptions are not confused with rover constraints.

☐ AI explanations are evidence-grounded and clearly separated from calculations.

☐ Core functionality works offline with cached data.

☐ Simulated, archival, inferred, and unavailable features are visibly distinguished.

☐ Tests, TypeScript checks, lint, and production build pass.

☐ README, reproducibility notes, licence, and AI-use disclosure are complete.

☐ The four-minute demo has been rehearsed offline.

13. Definition of Done

The project is competition-ready when its main Marswalk workflow works across all 10 supported regions, the evidence behind its scientific layers is traceable, estimates expose their assumptions, the core experience survives offline operation, and the team can demonstrate the system clearly in four minutes. A feature is not complete merely because it appears in the interface or has a passing unit test.

Build with evidence. Calculate deterministically. Explain transparently.